

A contiguous A5 block in the retained rational basis

Conditional multiplication saving $11447067/10^{12} > 2^{-17}$

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With substantial OpenAI Codex assistance

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Abstract

The prescribed common basis of icekylinx’s PR18 already contains a width-21 contiguous block inside an A5 corner previously charged as singleton pivots. We prove that ordered pivot pattern and combine it with Zhihao Chen’s PR21 finite construction and PR23 semantic/bulk transfer of RaD’s PR20 interfaces. The conditional multiplication saving is $\kappa = 11447067/10^{12} = 1.1447067 \times 10^{-5} > 2^{-17}$. No physical source frame, scalar producer, complex interface or maximum recursive width changes. Exact rational controls supplement the written argument; they are not formal verification of the inherited theorem.

1 Scope, dependencies and attribution

This contribution is one additional contiguous block in the retained rational basis and the resulting exact numerical assembly. The producer, positive labels, physical allocation and basis prescriptions are those of icekylinx’s PR18 [1]. The translated endpoints and complex phase interface are Zhihao Chen’s PR21 [2]. The semantic precision bound, bulk resampling and joint row-stock transfer are used exactly as composed by Zhihao Chen in PR23 [4], inheriting RaD/PR20 [3]. We retain the source-frame contributions of eumemic and the earlier contributions of Douglas Colkitt, Bortlesboat, dleen, icekylinx, Zhihao Chen and other authors recorded in NOTICE. Substantial assistance used in inherited work remains attributed there; this new contribution used OpenAI Codex assistance.

The result is conditional on the inherited analytic, Gaussian inverse, ordered-affine, fixed finite-alphabet multitape, and exact-recovery interfaces. The computational model and all uniformity hypotheses remain unchanged. This note is a focused extension; the complete retained proofs are preserved in the repository and cited below. No worldwide priority, practical speedup, linear-time or global optimality claim is made.

2 The new block

2.1 A contiguous block inside the retained A5 nullity corner

Keep the PR18 dimensions $(a, b, c) = (25, 23, 57)$, $H = 575$, the physical inner order $i = 23\alpha + \beta$, and all prescribed common-basis parameters. In particular, with $I = (0, \dots, 24)$ and $\Sigma = (0, \dots, 6)$, the first 82 row labels are $R = [I, \Sigma, I, I]$ and the last 82 inverse column labels are $C = [I, I, \Sigma, I]$.

After absorbing the free factors L_1, L_2 into the line coordinates, write the two rank-one inner projectors as $p\xi$ and $\nu\nu$, with $\xi p = \nu\nu = 1$. The inner A5 projector is $(I - p\xi) \otimes (I - \nu\nu)$, of rank

528 and nullity 47. Its null projector is $p\xi \otimes I + I \otimes v\nu - p\xi \otimes v\nu$. Index its first-47/last-47 corner by $0 \leq i, j < 47$, and put

$$\beta_i = i \bmod 23, \quad b_j = (528 + j) \bmod 23, \quad r_i = R_i, \quad c_j = C_{35+j}.$$

The prescribed rows of M_{β_i} and inverse columns of M_{b_j} give the exact corner formula

$$Q_{ij} = \mathbf{1}_{\beta_i=b_j} p r_i \xi_{c_j} + \mathbf{1}_{r_i=c_j} v_{\beta_i} \nu_{b_j} - p r_i \xi_{c_j} v_{\beta_i} \nu_{b_j}.$$

The first term uses the identity on the second factor; the second uses $M_{\beta_i} M_{b_j}^{-1}$ and hence the equality of the prescribed coefficient labels. The actual A5 corner is $-Q$. Its additional outer rank-one tensor factor only gives nonzero diagonal row and column rescalings in the physical first- H /last- H corner, which preserve ordered pivot positions.

Assume the line and dual coordinates occurring below are nonzero, as permitted by the same finite generic-coordinate conditions used in the retained basis proof. For $25 \leq j \leq 46$, one has $c_j = j - 22$ and $b_j = j - 24$. Thus the first ordered pivot is

$$Q_{0,46} = -p_0 \xi_{24} v_0 \nu_{22} \neq 0.$$

For $1 \leq i \leq 21$ and $25 \leq j \leq 45$, both Q_{0j} and $Q_{i,46}$ come entirely from the last rank-one term. Eliminating the first pivot therefore cancels that term exactly and leaves

$$\tilde{Q}_{ij} = \mathbf{1}_{i=j-24} p_i \xi_{j-22} + \mathbf{1}_{i=j-22} v_i \nu_{j-24}.$$

This is zero for $j > i + 24$, while $\tilde{Q}_{i,i+24} = p_i \xi_{i+2} \neq 0$. Inductively, the earlier pivot rows have no entries strictly to the right of their pivot columns, so eliminating them cannot change these entries. The rightmost-pivot lower/lower elimination consequently has pivots

$$(i, i + 24), \quad 1 \leq i \leq 21,$$

one increasing contiguous block of width 21. Diagonal pivot scalings are absorbed into the retained triangular factors. No gathering or reversal is introduced.

The existing primal and dual tree arguments still give full rank 47 for the nullity corner; the remaining 26 pivots may conservatively remain singleton children. The large-projector identity still gives the unchanged middle block of width 481. Therefore every A5 macro of the retained construction admits the profile

$26 \text{ singleton children; } 21, 481$

in place of 47 singleton children and 481. The rank sum is unchanged: $26 + 21 + 481 = 528$. For the $2N$ physical copies, subtract $21(2N)$ from the singleton multiplicity and add $2N$ width-21 children; the number of recursive calls decreases by $20(2N)$. All other profiles, physical roles, source frames, complex parameters and assumptions are unchanged. The new nonzero requirements are individual coordinates on the same rational parameter family, not additional structural restrictions; their finite product can be imposed together with the retained nonzero minors. The exact moment and final assembly with this profile are given in Section 4.

3 Compatibility with semantic precision and joint row stock

The regrouping changes the bit child multiset only. Each of the $2N$ A5 copies replaces 21 singleton children by one width-21 child. Thus the complete bit rank sum is conserved and the recursive call count falls by $40N$. In particular

$$m_b = 32775, \quad W_b = 5820554077800, \quad W_b m_b - s_b = 82076640800, \quad M_b = 32729$$

remain unchanged. The other children and physical streams are identical. The retained arbitrary-width interchange primitive implements the new contiguous block, and its triangular surrounding passes are already covered by the existing ordered-affine accounting.

The complex network is entirely unchanged. PR23's literal scalar charge, semantic induction, C_0 and $C_1 = 1$ therefore apply verbatim. Its bit and complex halving degrees depend only on (m_b, M_b) and (m_c, M_c) ; its wire-stock bounds depend only on W_b, W_c . All these quantities are unchanged. Consequently its degree-89000 joint row stock and restored nested bit factors remain valid. The new grouping requires no additional row family, padding operation, source translation or address reversal. This is the reason the PR23 bridge can be reused, rather than an inference from the numerical exponent alone.

4 Exact assembly

Theorem 1. *Under the retained interfaces just specified,*

$$T(n) = O\left(n(\log n)^{1-\kappa}\right), \quad \kappa = \frac{11447067}{10^{12}} > 2^{-17}.$$

Proof. The new exact bit child multiset gives

$$a = \frac{1144733}{10^{11}}, \quad \frac{\sum_t c_t t^{1-a}}{W_b m_b^{1-a}} < 1 - \frac{3}{10^{14}}.$$

The certificate uses the retained rational logarithm enclosures and $\exp(u) \leq 1 + u + u^2/[2(1 - u/3)]$ in its stated range. The next 10^{-12} grid point does not pass this upper-enclosure certificate; this is not a lower bound on the actual moment. Omitting the new A5 block fails to certify the displayed a . The unchanged complex saving is $b = 18/10^6$.

Use the PR23 parameter formulas with

$$\begin{aligned} h &= 10^{-8}, \quad \beta = 1/4, \quad \delta = h/8, \quad \tau = 1 - a, \quad \sigma = 1 - b, \quad q = a(1 - 2h), \\ \lambda' &= 1 - q, \quad \lambda = (\tau + \lambda')/2, \quad c = q(1 + h), \quad \epsilon = \frac{1 - h}{1 + c + q}, \\ G &= \epsilon q, \quad r = (G + 1 - \epsilon)/2, \quad C_1 = 1. \end{aligned}$$

The complex leaf saving $(1 - \beta)b = 13.5 \times 10^{-6}$ remains larger than a . All 47 strict assembly constraints pass exact rational arithmetic, including the literal scalar charge and joint row-stock gap. The seven margins are

$$\begin{aligned} g_1 &= 1 - \epsilon(1 + c), & g_2 &= a, & g_3 &= G, & g_4 &= a, \\ g_5 &= \min(1 - \epsilon - \delta, r - \delta), & g_6 &= 1 - \epsilon - \delta, & g_7 &= \epsilon. \end{aligned}$$

Their minimum is

$$G = \frac{520333166208186467703}{45455586120893480913532297}, \quad G - \kappa > \frac{5}{10^{13}} > 0.$$

The positive slack supplies the retained strict logarithmic absorption. The PR23 eventual-cutoff calculation, reevaluated at these parameters, remains valid with the same fixed interfaces and row-stock degree. $\square$

5 Reproduction and verification boundary

Run `python3 research/a5-semantic/witness.py` for the new rational bit moment, pinned-input hashes, exact corner controls, unchanged bridge, 47 constraints, seven margins and cutoff arithmetic. The four exact rational corner controls check both signs and every stated Schur entry. They reject omission of the first pivot cancellation and alteration of an inverse-column boundary. General simultaneous validity follows from the written identity and retained nonzero-minor argument, not from a finite sample. The inherited complete producer rebuilds remain retained source-checked evidence; `make verify` performs the repository rerun. The focused patch is against pinned PR23 source, not the original OpenAI upstream manuscript. Its exact base is recorded in `research/a5-semantic/SOURCE.json`.

References

- [1] icekylinx. Partial-swap construction and prescribed common basis. <https://github.com/CrocSwap/integer-mult-bounds/pull/18>.
- [2] Zhihao Chen (jacklightChen). Translated partial-swap and phase frames. <https://github.com/CrocSwap/integer-mult-bounds/pull/21>.
- [3] RaD. Semantic precision and bulk resampling interfaces. <https://github.com/CrocSwap/integer-mult-bounds/pull/20>.
- [4] Zhihao Chen (jacklightChen). PR21 with semantic precision and bulk transfer. <https://github.com/CrocSwap/integer-mult-bounds/pull/23>.